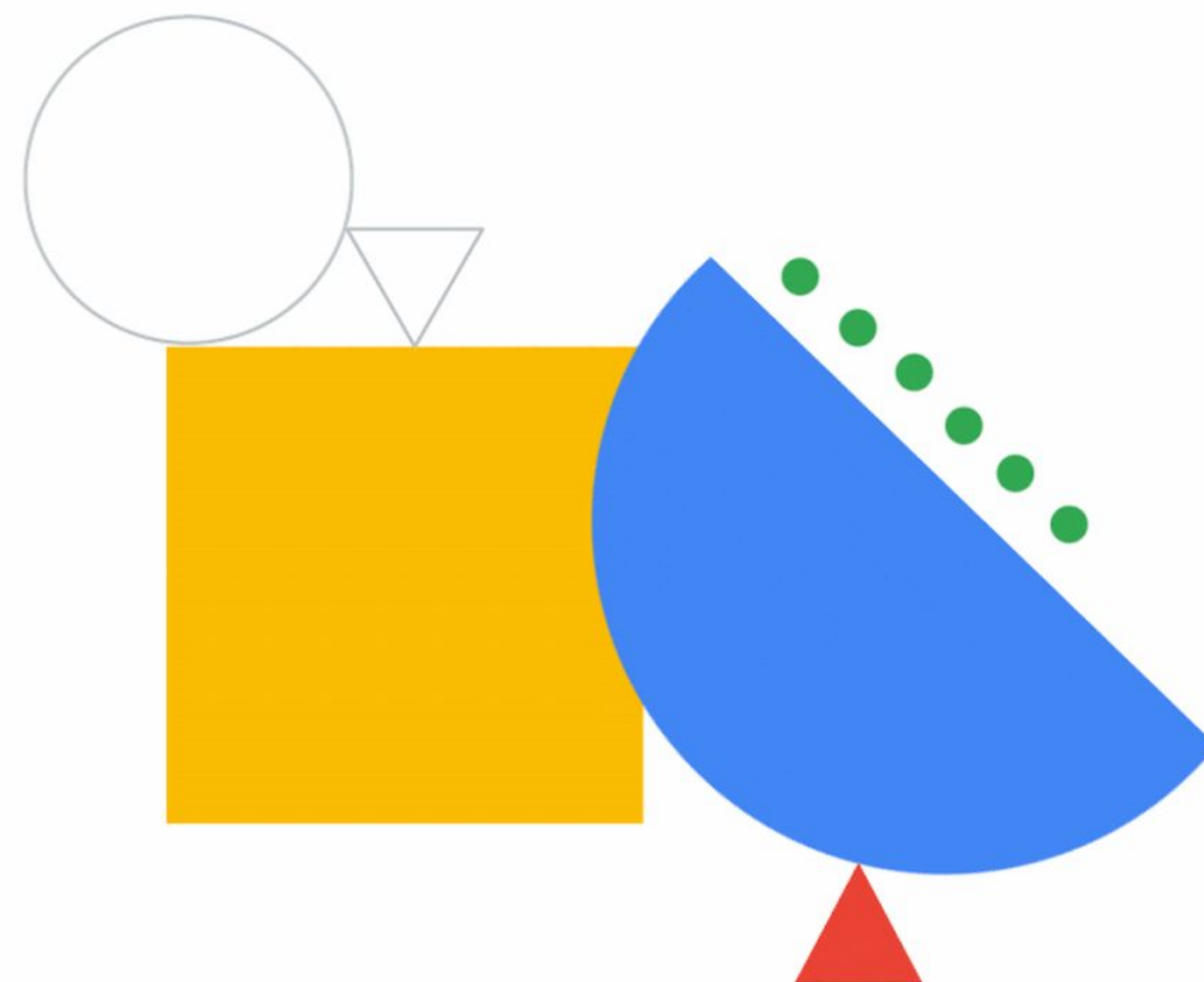


# Introduction to Google Cloud Observability



# Objectives

- 01 Describe the purpose and capabilities of Google Cloud Observability
- 02 Explain the purpose of Cloud Monitoring
- 03 Explain the purpose of Cloud Logging and Error Reporting
- 04 Explain the purpose of Cloud Trace

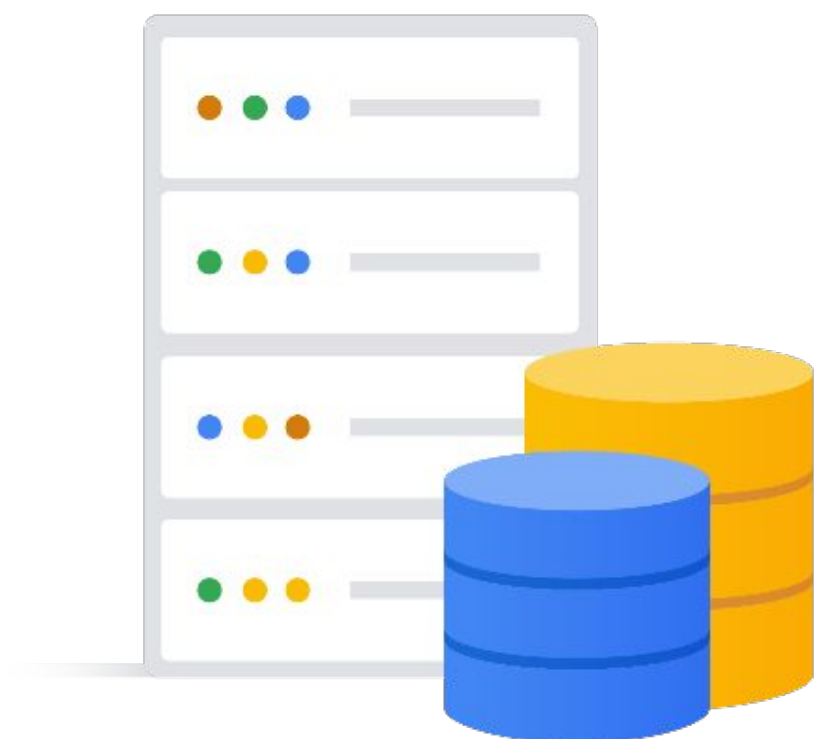


# In this section, you explore



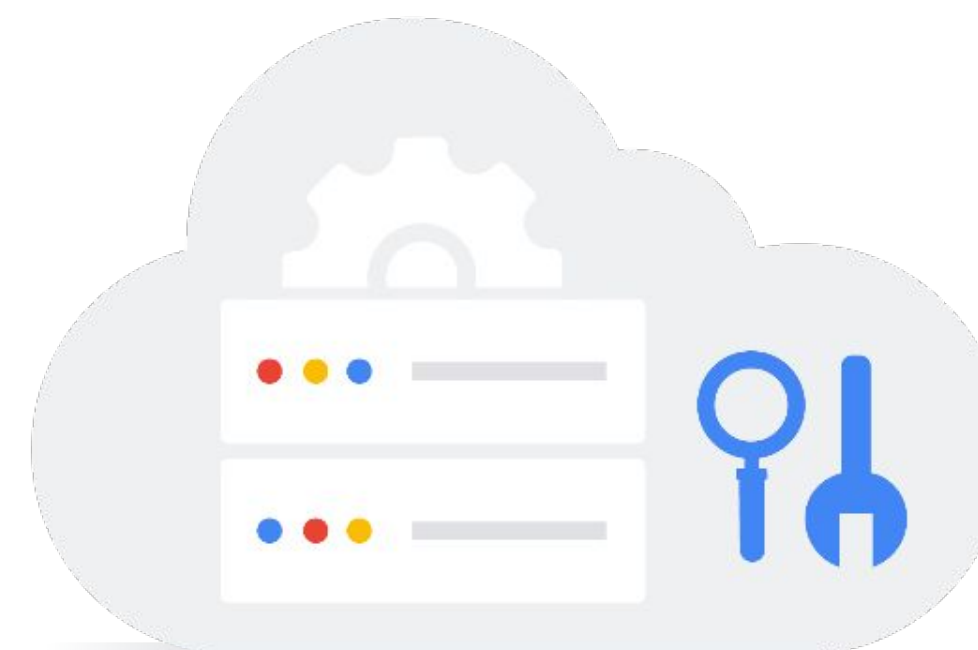
- ✓ Need for Google Cloud observability
- ✓ Cloud Monitoring
- ✓ Cloud Logging
- ✓ Error Reporting
- ✓ Application Performance Management

## On-premises



Physical check

## Cloud



Observability tools

# Need for observability

## Visibility into system health

---

Help me understand my application and tell me if it's healthy

## Error reporting and alerting

---

Bring my attention directly to problems

## Efficient troubleshooting

---

Help me fix it if it's broken

## Improve performance

---

Guide me to optimize it

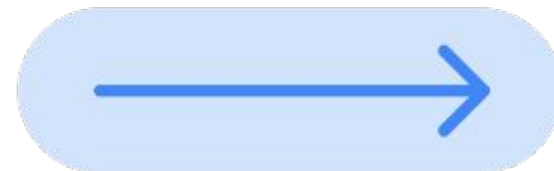


# Monitoring gives you real-time system information



Google's *Site Reliability Engineering* book

[landing.google.com/sre/books](https://landing.google.com/sre/books)



Collecting, processing, aggregating, and displaying **real-time quantitative data about a system**, such as:

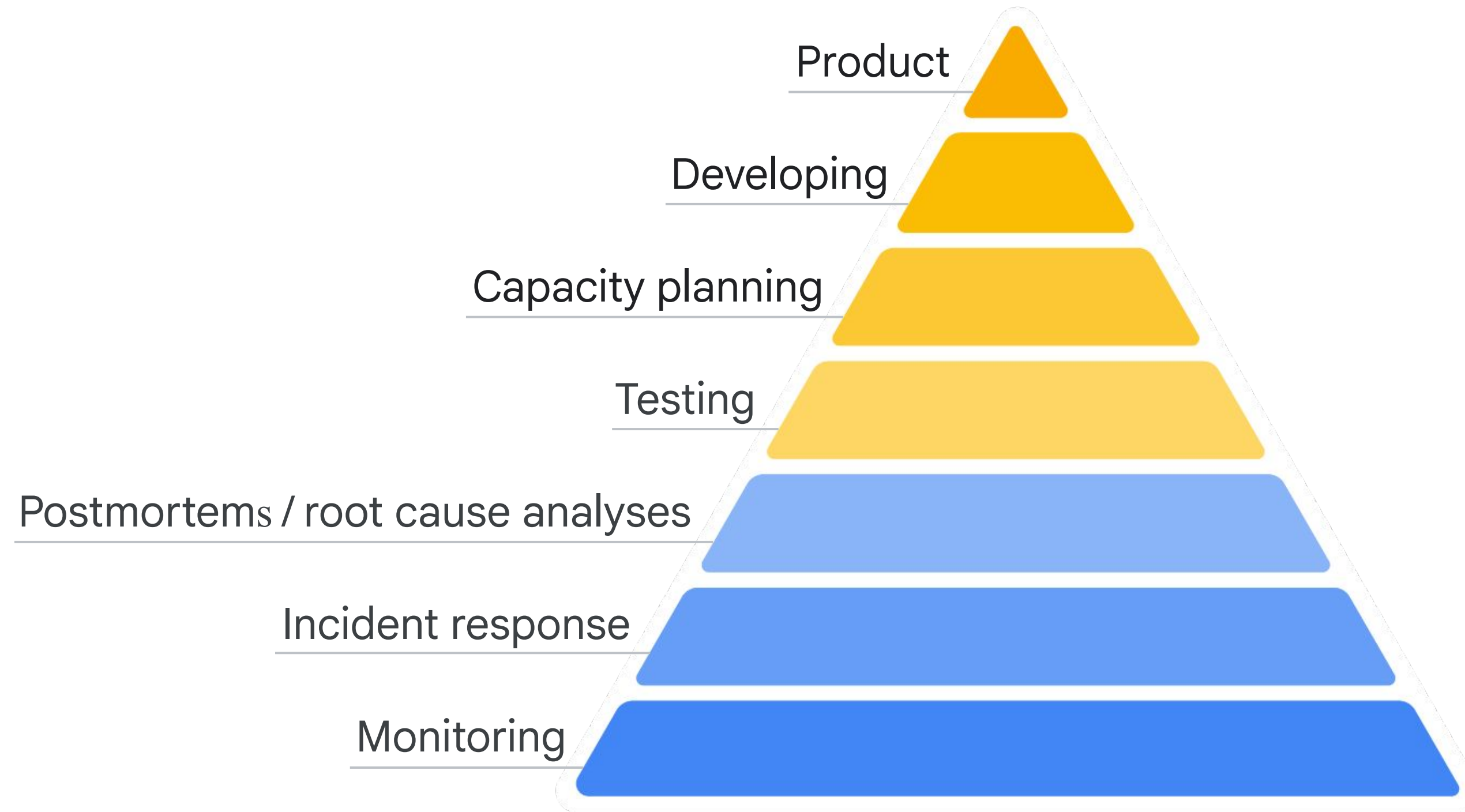
Query counts and types

Error counts and types

Processing times

Server lifetimes

# Monitoring gives you real-time system information



# What's needed from products



Continual  
improvement



Dashboards



Automated  
alerts

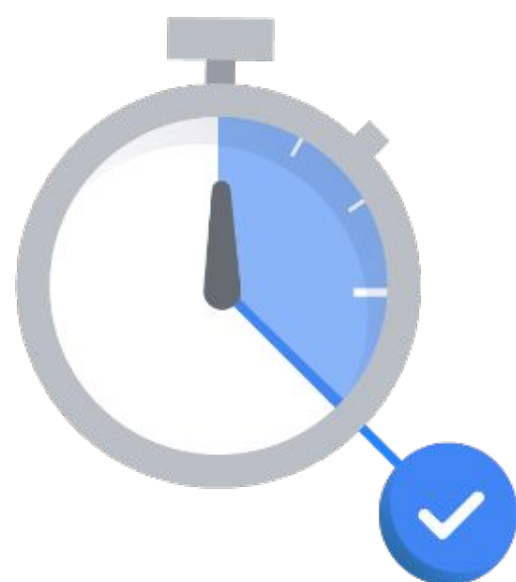


Incident response



# Four golden signals

Latency



Traffic



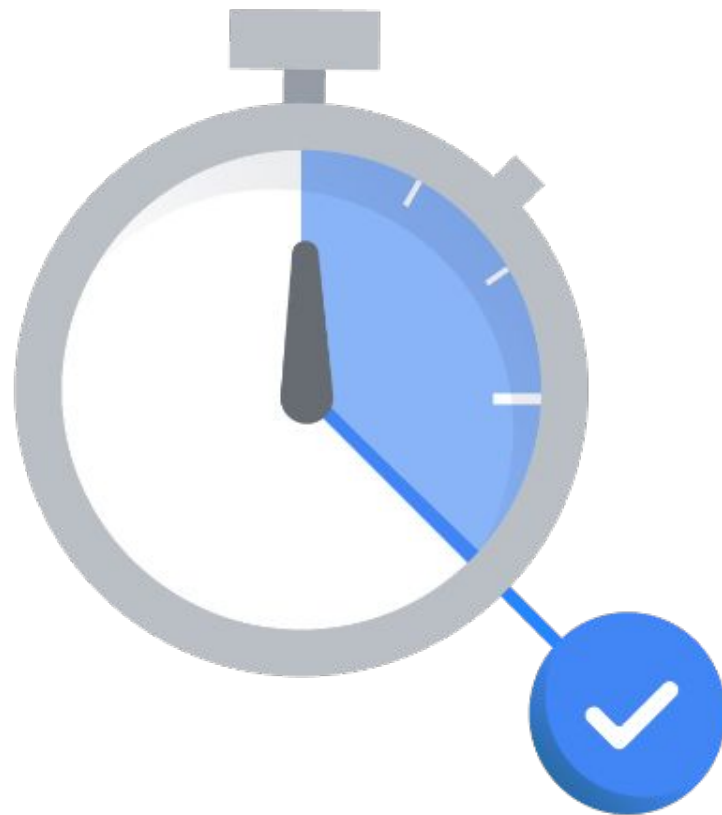
Saturation



Errors



# The importance of latency



01

Changes in latency could indicate emerging issues.

02

Its values may be tied to capacity demands.

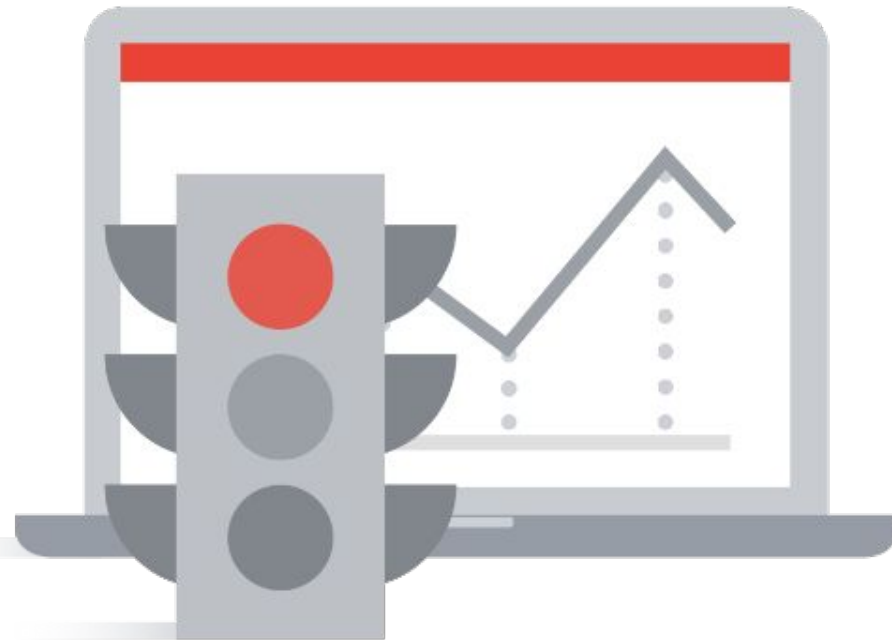
03

It can be used to measure system improvements.



- Page load latency
- Number of requests waiting for a thread

# The importance of traffic



01

It's an indicator of current system demand.

02

Its historical trends are used for capacity planning.

03

It's a core measure when calculating infrastructure spend.



- # retrievals per second
- # active requests

# The importance of saturation



01

It's an indicator of how full the service is.

02

It focuses on the most constrained resources.

03

It's frequently tied to degrading performance as capacity is reached.



- % memory utilization
- % thread pool utilization

# The importance of errors



01

They may indicate configuration or capacity issues

02

They can indicate service level objective violations

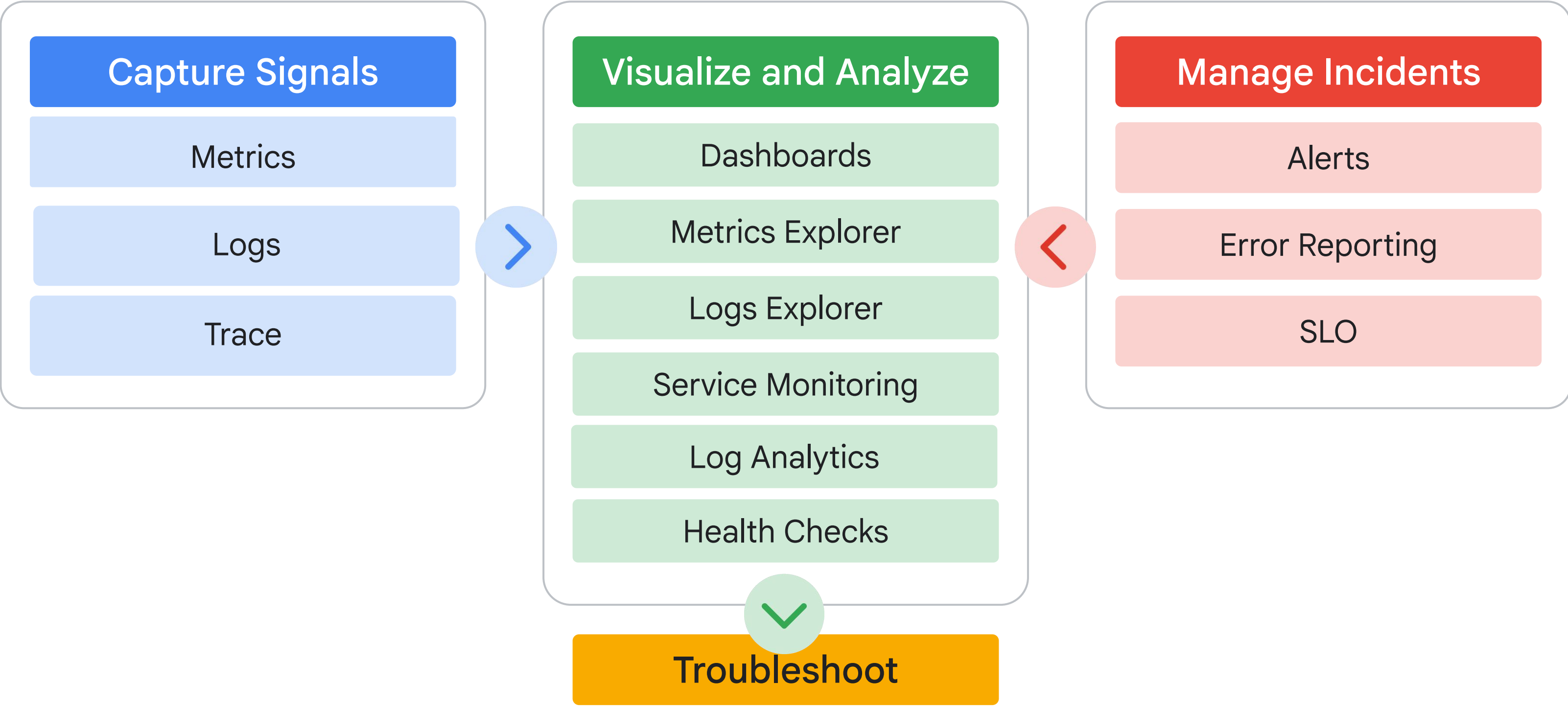
03

An error might mean it's time to send out an alert



- # failed requests
- # exceptions





# Google Cloud Observability

Cloud  
Monitoring



Cloud  
Logging



Error  
Reporting



Cloud  
Trace



Cloud  
Profiler



# In this section, you explore



- ✓ Need for Google Cloud Observability
- ✓ Cloud Monitoring
- ✓ Cloud Logging
- ✓ Error Reporting
- ✓ Application Performance Management





# Cloud Monitoring

- ✓ Provides visibility into the performance, uptime, and overall health of cloud-powered applications.
- ✓ Collects metrics, events, and metadata from projects, logs, services, systems, agents, custom code, and various common application components.
- ✓ Ingests that data and generates insights via dashboards, Metrics Explorer charts, and automated alerts.

# Cloud Monitoring features



## Many free metrics

On 100+ monitored resources services, over 1500+ metrics are immediately available with no cost



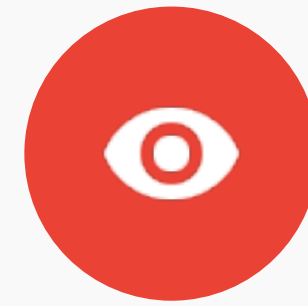
## Open source standards

Leverage Prometheus and Open Telemetry to collect metrics across compute workloads



## Customization for key workloads

Tune with Managed Service for Prometheus on GKE and Cloud Ops Agent on Compute Engine



## In-context visualizations & alerts

View relevant telemetry data alongside your workloads across Google Cloud

# In this section, you explore



- ✓ Need for Google Cloud Observability
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# Cloud Logging

- ✓ Allows users to collect, store, search, analyze, monitor, and alert on log entries and events.
- ✓ Provides automatic ingestion with simple controls for routing, storing, and displaying your log data.
- ✓ Leverage tools like Log Analytics to view trends, or Error Reporting and Log Explorer to quickly examine problems.

# Logging has multiple aspects

## Collect

- Cloud events, configuration changes, and from customer services
- Logs at various level of the resource hierarchy

## Analyze

- Log data in real time with the integrated Logs Explorer
- Run queries and analyze with Log Analytics
- Exported logs from Cloud Storage or BigQuery

## Export

- Export to Cloud Storage, or Pub/Sub, or BigQuery
- Logs-based metrics for augmented Monitoring

## Retain

- Data access and service logs for 30 days and admin logs for 400 days
- Longer-term in Cloud Storage or BigQuery



# Developers use cases



## Developers

- Troubleshooting
- Debugging

**Get started quickly** – Out-of-the-box collection of system metrics and logs

**Use logging SDKs and library** – Integration into popular SDKs to support rich log formatting

**Analyze log in real-time** – Analyze log data in real-time, debug code, troubleshoot your apps

**Find errors quickly** – Find errors via stack traces automatically with Error Reporting

# Operators use cases



## Operators

- SLO/alerting
- Log management
- Workload management
- Cost management

**Collect the right telemetry** – Instrumentation for Compute Engine, on-prem and other cloud providers

**Centralize logs** – Centralize logs for specific users, teams and/or organizations

**Manage logs** – Set retention periods, select supported regions for regional data storage

**Set alerts** – Understand log volume/cost, set alerts on important application metrics

**Export logs** – Export to Google Cloud for storage, analysis, integrate with 3rd parties

# Security operations use cases



## SecOps Analyst

Primarily concerned with secure operations of the Google Cloud fleet of resources. This persona uses the platform features to meet organizational security requirements.

**Collect audit logs** – Collect Google Cloud audit logs by default, advanced security logs such as data access logs

**Collect network telemetry data** – Collect and analyze VPC flow logs, GKE network, firewall, load balancer logs

**Analyze logs for security events** – View audit logs and other events to investigate possible security events



# In this section, you explore



- ✓ Need for Google Cloud Observability
- ✓ Cloud Monitoring
- ✓ Cloud Logging
- ✓ **Error Reporting**
- ✓ Application Performance Management



## Error Reporting

Error Reporting **identifies, counts, analyzes, and aggregates** the crashes in your running cloud services.

# Error Reporting features

## Real time processing

---

Applications errors are processed and displayed within seconds

## Quickly view and understand errors

---

A dedicated page displays the details of the error

## Instant notification

---

You are notified when events occur

# Error Reporting interface

## Errors in the last 30 days

| Resolution Status                      | Occurrences <span>▼</span>                                                         | Error                                                                                                                                           | Seen in       |
|----------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <span>✓</span> Resolved <span>▼</span> |  | <b>NEW</b> <b>PermissionDenied: 403 The caller does not have permission</b><br>raise_from (/usr/lib/python2.7/dist-packages/six.py)             | gke_instances |
| <span>!</span> Open <span>▼</span>     |  | <b>NEW</b> <b>ServiceUnavailable: 503 Getting metadata from plugin failed with erro</b><br>raise_from (/usr/lib/python2.7/dist-packages/six.py) | gke_instances |
|                                        |                                                                                    |                                                                                                                                                 | gke_instances |

## Stack trace sample

Parsed   Raw

```
PermissionDenied: 403 The caller does not have permission
  at raise_from (/usr/lib/python2.7/dist-packages/six.py:737)
  at error_remapped_callable (/usr/local/lib/python2.7/dist-packages/google/api_core/grpc_helpers.py:56)
  at __call__ (/usr/local/lib/python2.7/dist-packages/google/api_core/gapic_v1/method.py:139)
  at batch_write_spans (/usr/local/lib/python2.7/dist-packages/google/cloud/trace_v2/gapic/trace_service_client.py:18)
```

gke\_instances

gke\_instances

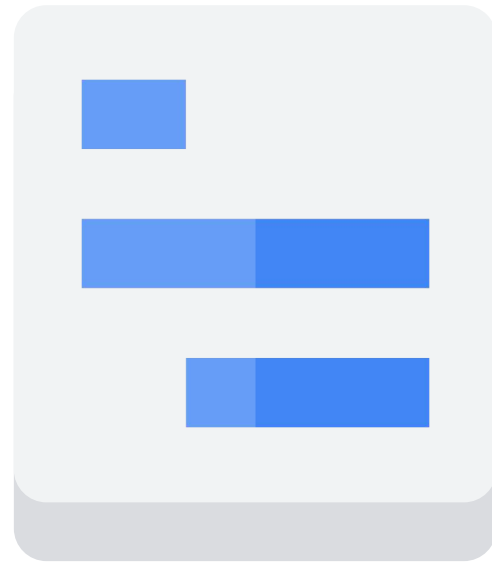
gke\_instances

# In this section, you explore



- ✓ Need for Google Cloud Observability
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# Cloud Trace



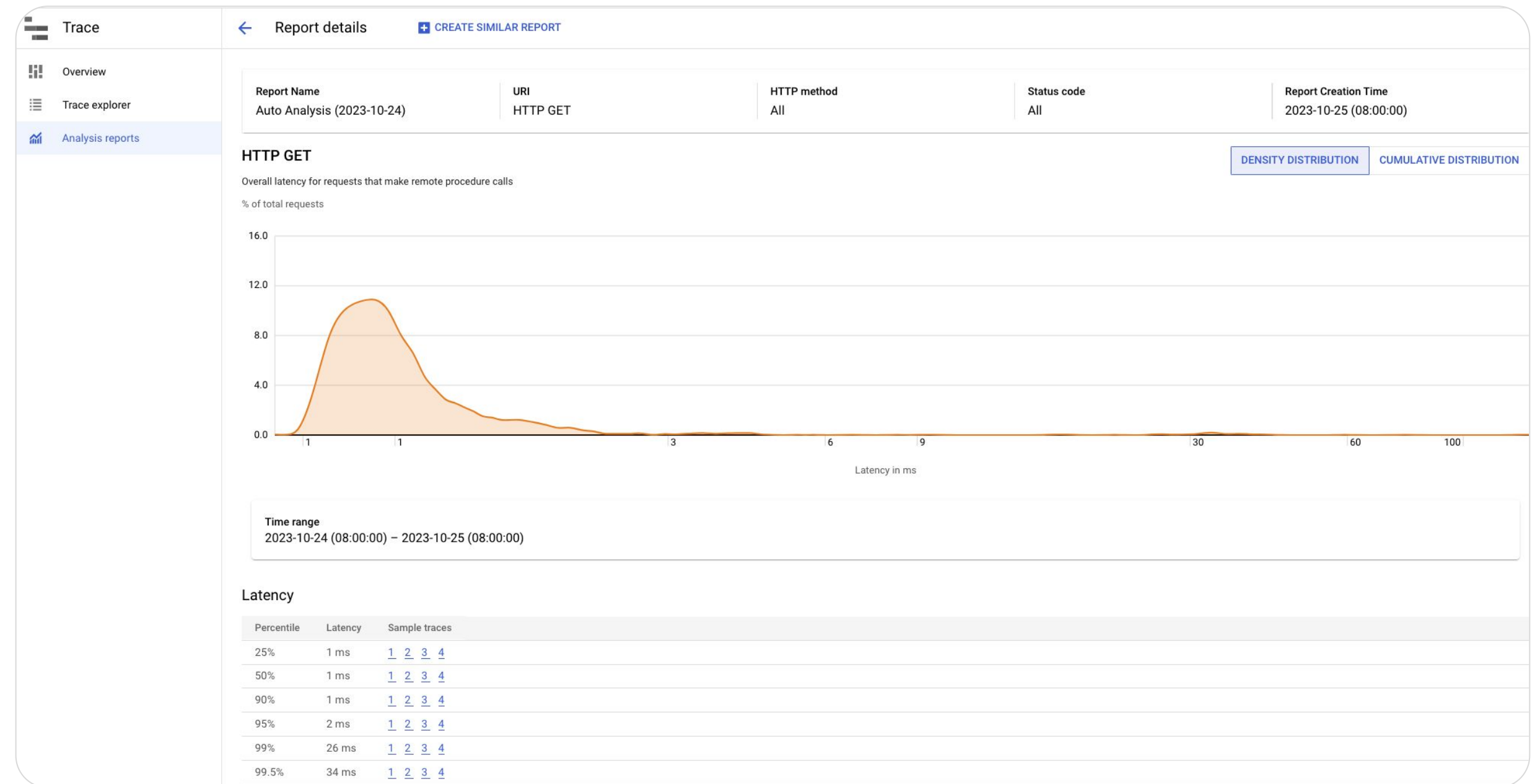
Collects latency data from distributed applications and displays it in the Google Cloud console.



Captures traces from applications deployed on App Engine flexible and standard environment, Compute Engine VMs, Google Kubernetes Engine containers, Cloud Run and non-Google Cloud environments.

# Latency reports

- Provide performance insights in near-real time.
- Generate in-depth latency reports to surface performance degradations.
- Identify recent changes to application performance.







# Cloud Profiler

- ✓ Uses statistical techniques and extremely low-impact instrumentation to provide a complete picture of an application.
- ✓ Allows developers to analyze applications running anywhere.
- ✓ Presents the call hierarchy and resource consumption of the relevant function in an interactive flame graph.



# Google Cloud Observability helps you explore the known and unknown issues

## User-focused products

Understand a customer's journey with SLO monitoring, uptime checks, tracing and more.

## Open, flexible foundations

Leverage popular open source projects like Prometheus, OpenTelemetry, and Fluentbit.

## Integrated for ease

Automatically ingest log, connect data sets, collect in-context telemetry across Google Cloud service.

## Meaningful analysis and alerting

Use powerful analysis tools and leverage alerting for both automated and human-led resolutions.



# Knowledge Check



# Quiz | Question 1

## Question

You want a simple way to see the latency of requests for a web application you deployed to Cloud Run. What Google Cloud tool should you use?

- A. Trace
- B. Profiler
- C. Metrics Explorer
- D. Logs Explorer

# Quiz | Question 1

## Answer

You want a simple way to see the latency of requests for a web application you deployed to Cloud Run. What Google Cloud tool should you use?

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- D. Logs Explorer



# Quiz | Question 2

## Question

You want to examine messages generated by running code. Which tool might be best for doing this?

- A. Trace
- B. Profiler
- C. Metrics Explorer
- D. Logs Explorer



# Quiz | Question 2

## Answer

You want to examine messages generated by running code. Which tool might be best for doing this?

- A. Trace
- B. Profiler
- C. Metrics Explorer
- D. Logs Explorer



# Quiz | Question 3

## Question

Users have reported that an application occasionally returns garbage data instead of the intended results, but you have been unable to reproduce this problem in your test environment. Which tool might be of best help?

- A. Trace
- B. Profiler
- C. Error Reporting
- D. Logs Explorer

# Quiz | Question 3

## Answer

Users have reported that an application occasionally returns garbage data instead of the intended results, but you have been unable to reproduce this problem in your test environment. Which tool might be of best help?

- A. Trace
- B. Profiler
- C. Error Reporting
- D. Logs Explorer





# Quiz | Question 4

## Question

You want to calculate the uptime of a service and receive alerts if the uptime value falls below a certain threshold. Which tool will help you with this requirement?

- A. Cloud Monitoring
- B. Profiler
- C. Error Reporting
- D. Logs Explorer

# Quiz | Question 4

## Question

You want to calculate the uptime of a service and receive alerts if the uptime value falls below a certain threshold. Which tool will help you with this requirement?

- A. Cloud Monitoring
- B. Profiler
- C. Error Reporting
- D. Logs Explorer



# Recap

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